

Breaking voices

During puberty, the voices of both males and females change to a lower pitch than that of pre-puberty. The deepening in females' voices is hardly noticeable, but some males' voices drop dramatically. When this happens, people say a male's voice is "breaking".

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How the voice works

The larynx, often known as the "voice box," combines with a system of cavities in the face and throat to enable people to manipulate the sounds they make to talk and sing.

When air is pushed out of the lungs, it rushes up the trachea, or "windpipe," and through the larynx. Stretched across the larynx are two vocal ligaments, or "vocal cords," which are a bit like elastic bands. They vibrate when air passes between them—much like when a guitar string is strummed—and produce the sound of the voice.

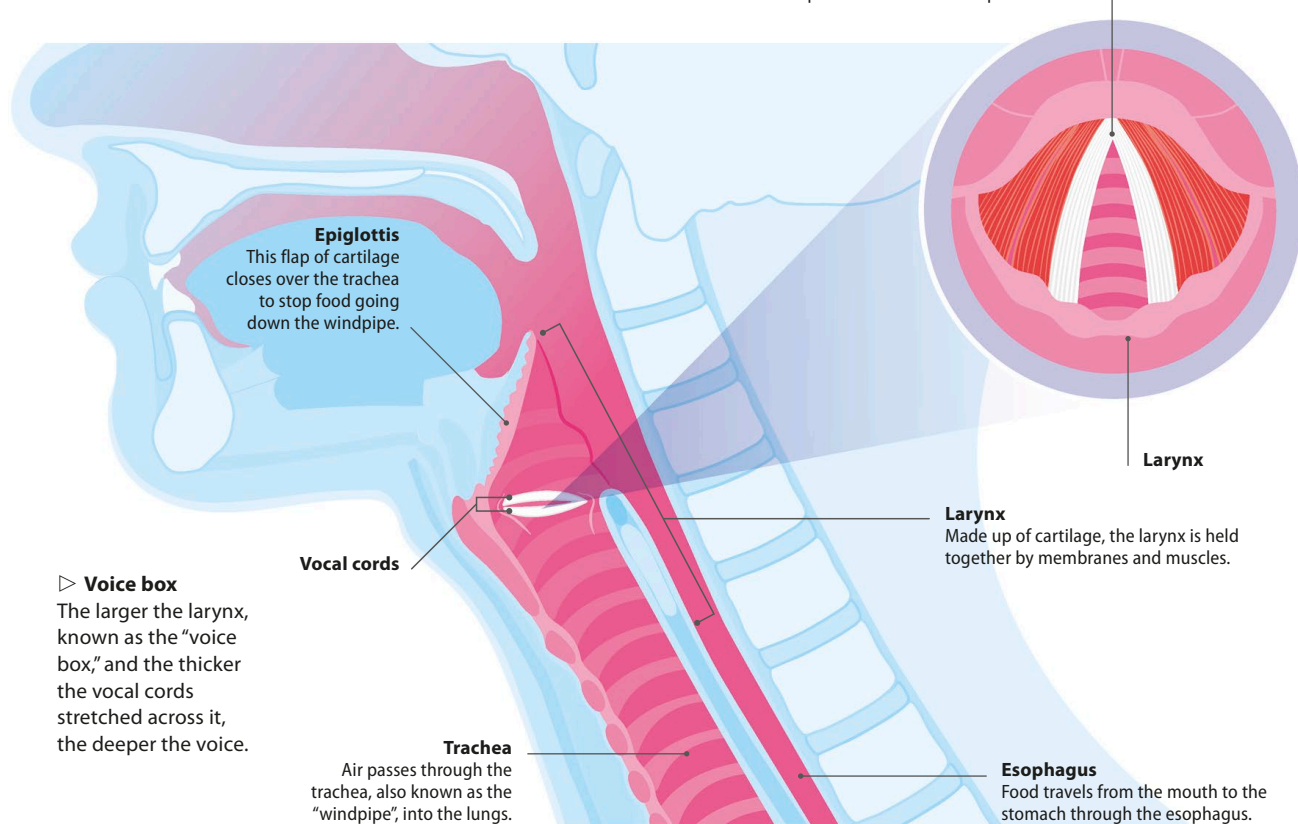
The impact of puberty

During puberty, testosterone makes the cartilage in the voice box grow. The vocal cords become 60 percent longer and thicker, and so vibrate at a lower frequency than before—reducing from 200 times per second to as low as 130 times per second. This makes the voice sound much deeper.

It's not clear why this change to the voice occurs. In other animals, males develop deeper voices to attract a mate and to frighten off predators and competing males more effectively. It may be that the human voice changes for the same reasons.

Vocal cords

The V-shape vocal cords, which stretch across the larynx, are "strummed" by air rushing past them to produce the sound of a person's voice.



▷ Voice box

The larger the larynx, known as the "voice box," and the thicker the vocal cords stretched across it, the deeper the voice.